



**Kampus
Merdeka**
INDONESIA JAYA

RevoU Tech Academy **Data Analytics** **Capstone Project**

Prepared and Presented by : Team 7 Section Sabang



Kampus
Merdeka
INDONESIA JAYA

US Candy Distributor

Performance Report

Prepared and Presented by : Team 7 Section Sabang

OUR TEAM MATE'S

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Executive Summary

- 01 Project Description
- 02 Objectives
- 03 Tools
- 04 Metrics
- 05 Analysis Result
- 06 Recommendation



- **Sales:** Contains information on product sales across various cities.
- **Factory:** Includes details about the locations and operations of production facilities.
- **Product:** Provides information on the marketed product details.
- **Target:** Indicates product divisions along with their sales targets.
- **US Zips:** Contains ZIP code data for the United States, assisting in identifying geographical locations.

Dataset US Candy Distributor Team 7

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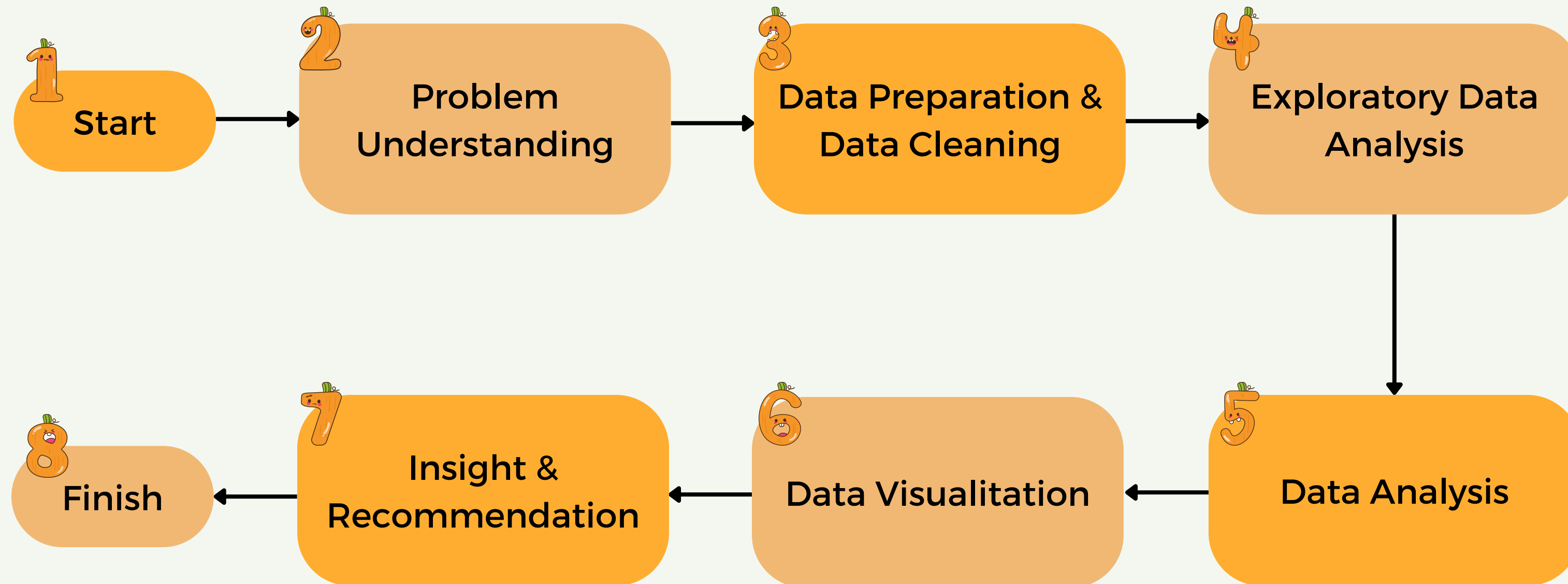
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	A	B	C	D	E	F	G	H	I	J	K	L
1	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Country/Region	City	State/Province	Postal Code	latitude	longitude
2	282	US-2021-128055-CHO-TRI-54000	31/3/2021	26/9/2026	Standard Class	128055	United States	San Francisco	California	94122	3,776,113	-12,248,433
3	288	US-2021-128055-CHO-SCR-58000	31/3/2021	26/9/2026	Standard Class	128055	United States	San Francisco	California	94122	3,776,113	-12,248,433
4	1132	US-2021-138100-CHO-FUD-51000	15/9/2021	13/3/2027	Standard Class	138100	United States	New York City	New York	10011	4,074,173	-7,400,037
5	1133	US-2021-138100-CHO-MIL-31000	15/9/2021	13/3/2027	Standard Class	138100	United States	New York City	New York	10011	4,074,173	-7,400,037
6	3396	US-2022-121391-CHO-MIL-31000	4/10/2022	29/3/2028	First Class	121391	United States	San Francisco	California	94109	3,779,334	-12,242,138
7	4370	US-2023-103982-CHO-SCR-58000	3/3/2023	28/8/2028	Standard Class	103982	United States	Round Rock	Texas	78664	3,050,461	-9,764,583
8	4373	US-2023-103982-CHO-NUT-13000	3/3/2023	28/8/2028	Standard Class	103982	United States	Round Rock	Texas	78664	3,050,461	-9,764,583
9	4375	US-2023-103982-CHO-FUD-51000	3/3/2023	28/8/2028	Standard Class	103982	United States	Round Rock	Texas	78664	3,050,461	-9,764,583
10	4377	US-2023-103982-CHO-MIL-31000	3/3/2023	28/8/2028	Standard Class	103982	United States	Round Rock	Texas	78664	3,050,461	-9,764,583
11	8005	US-2024-147039-CHO-NUT-13000	29/6/2024	25/12/2029	Standard Class	147039	United States	Minneapolis	Minnesota	55407	4,493,484	-9,325,263
12	8009	US-2024-147039-CHO-NUT-13000	29/6/2024	25/12/2029	Standard Class	147039	United States	Minneapolis	Minnesota	55407	4,493,484	-9,325,263
13	387	US-2021-158064-CHO-MIL-31000	21/4/2021	16/10/2026	Standard Class	158064	United States	Los Angeles	California	90008	3,401,037	-1,183,428
14	1397	US-2021-130729-CHO-MIL-31000	24/10/2021	21/4/2027	Standard Class	130729	United States	Rancho Cucamonga	California	91730	3,410,023	-1,175,811
15	2117	US-2022-140921-CHO-TRI-54000	3/2/2022	29/7/2027	First Class	140921	United States	Omaha	Nebraska	68104	4,129,525	-9,600,104
16	2121	US-2022-140921-CHO-NUT-13000	3/2/2022	29/7/2027	First Class	140921	United States	Omaha	Nebraska	68104	4,129,525	-9,600,104
17	2518	US-2022-109939-CHO-SCR-58000	8/5/2022	2/11/2027	Standard Class	109939	United States	Salem	Oregon	97301	4,494,948	-12,300,439
18	3660	US-2022-114503-CHO-NUT-13000	13/11/2022	9/5/2028	Standard Class	114503	United States	Lebanon	Tennessee	37087	362,733	-8,627,224
19	5189	US-2023-126613-CHO-TRI-54000	10/7/2023	5/1/2029	Standard Class	126613	United States	Mesa	Arizona	85204	3,339,666	-11,178,786
20	6266	US-2023-131065-CHO-NUT-13000	14/11/2023	8/5/2029	Second Class	131065	United States	Atlanta	Georgia	30318	3,379,235	-8,444,831
21	6269	US-2023-131065-CHO-SCR-58000	14/11/2023	8/5/2029	Second Class	131065	United States	Atlanta	Georgia	30318	3,379,235	-8,444,831
22	6271	US-2023-131065-CHO-NUT-13000	14/11/2023	8/5/2029	Second Class	131065	United States	Atlanta	Georgia	30318	3,379,235	-8,444,831
23	8580	US-2024-169488-CHO-MIL-31000	7/9/2024	2/3/2030	First Class	169488	United States	Providence	Rhode Island	2908	4,183,877	-7,143,926
24	8582	US-2024-169488-CHO-TRI-54000	7/9/2024	2/3/2030	First Class	169488	United States	Providence	Rhode Island	2908	4,183,877	-7,143,926
25	9950	US-2024-100230-CHO-NUT-13000	11/12/2024	7/6/2030	Standard Class	100230	United States	New York City	New York	10035	4,079,538	-7,392,965
26	9956	US-2024-100230-CHO-TRI-54000	11/12/2024	7/6/2030	Standard Class	100230	United States	New York City	New York	10035	4,079,538	-7,392,965
27	9961	US-2024-100230-CHO-NUT-13000	11/12/2024	7/6/2030	Standard Class	100230	United States	New York City	New York	10035	4,079,538	-7,392,965
28	430	US-2021-155271-CHO-TRI-54000	4/5/2021	25/10/2026	Same Day	155271	United States	Middletown	Connecticut	6457	4,154,759	-7,265,484
29	5231	US-2023-121671-CHO-TRI-54000	17/7/2023	11/1/2029	Standard Class	121671	United States	Springfield	Missouri	65807	37,166	-9,332,483
30	5232	US-2023-121671-CHO-SCR-58000	17/7/2023	11/1/2029	Standard Class	121671	United States	Springfield	Missouri	65807	37,166	-9,332,483
31	5233	US-2023-121671-CHO-NUT-13000	17/7/2023	11/1/2029	Standard Class	121671	United States	Springfield	Missouri	65807	37,166	-9,332,483
32	5235	US-2023-121671-CHO-NUT-13000	17/7/2023	11/1/2029	Standard Class	121671	United States	Springfield	Missouri	65807	37,166	-9,332,483
33	5237	US-2023-121671-CHO-FUD-51000	17/7/2023	11/1/2029	Standard Class	121671	United States	Springfield	Missouri	65807	37,166	-9,332,483
34	5462	US-2023-114601-CHO-SCR-58000	26/8/2023	22/2/2029	Standard Class	114601	United States	Detroit	Michigan	48234	424,312	-8,303,952
35	5464	US-2023-114601-CHO-SCR-58000	26/8/2023	22/2/2029	Standard Class	114601	United States	Detroit	Michigan	48234	424,312	-8,303,952
36	5466	US-2023-114601-CHO-NUT-13000	26/8/2023	22/2/2029	Standard Class	114601	United States	Detroit	Michigan	48234	424,312	-8,303,952

Methodology



Business Analysis

Business Background

In US candy distribution industry, **distribution efficiency** plays a crucial role in enhancing profitability and customer satisfaction. By leveraging historical sales data and geographical information such as **latitude** and **longitude**, companies can identify sales patterns with the highest sales contributions.

Problem Statement

How to **increase distribution efficiency** to **cities** using **latitude** and **longitude** data of **3-5 areas** with highest sales contributing **10% of total sales within 1 year?**

Business Analysis

Objective of Analysis

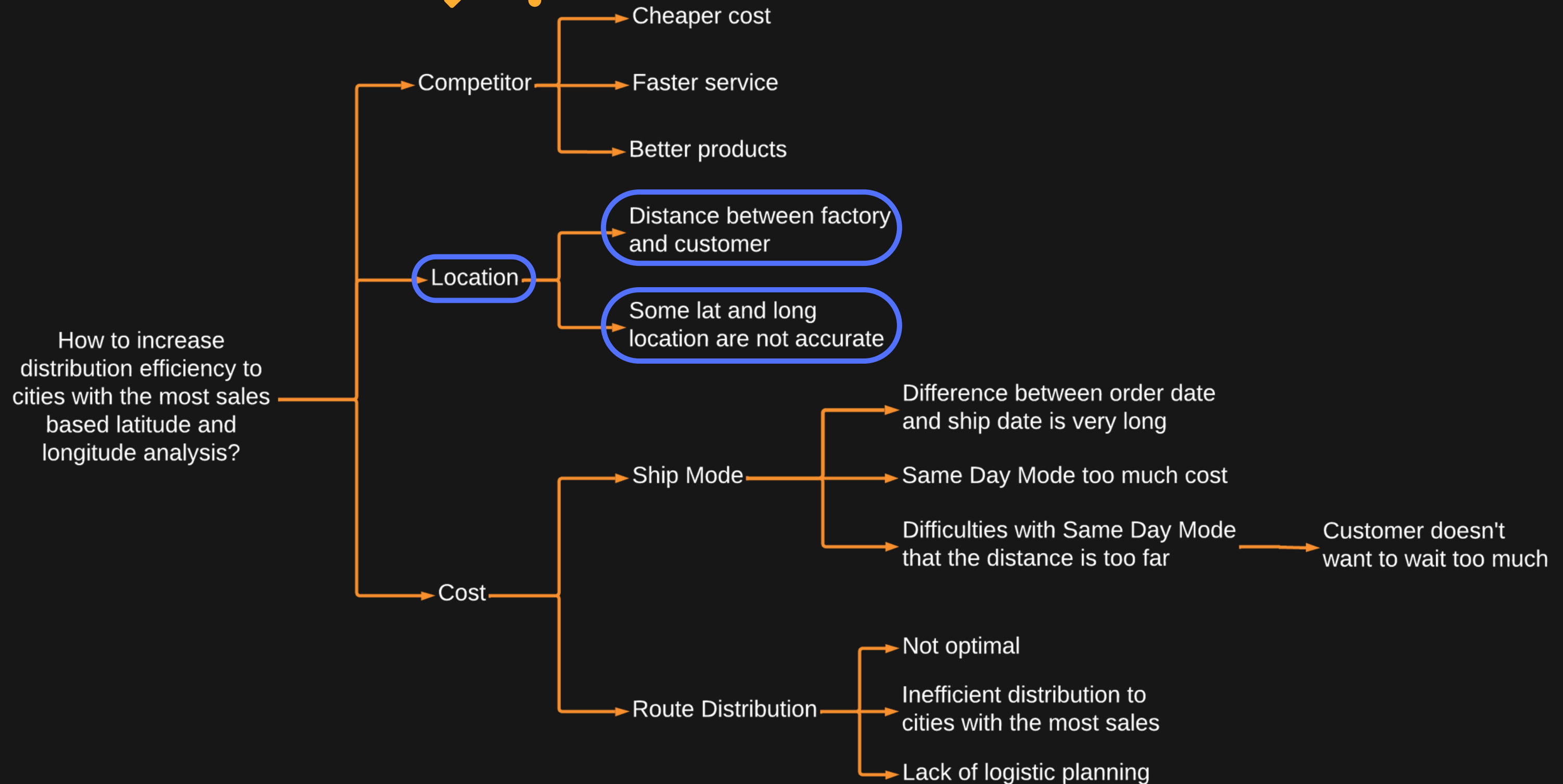
To **increase the distribution efficiency** to **cities** using **latitude** and **longitude** data of **3-5 areas** with highest sales contributing **10% of total sales within 1 year**.

Possible Cause

1. **Geographical Distance:** The distance between cities with the highest sales and the distribution factory locations is too far, leading to increased distribution time and costs.
2. **Disparity in Factory Location:** The production facilities are unevenly distributed or overly concentrated in specific regions, making it inefficient to serve areas with high sales.
3. **High Customer Demand in Distant Areas:** Cities with the highest sales demand are located far from the main factory or warehouse.

Business Analysis

Root Cause Analysis



Business Analysis

Metrics

- **Total Sales per City** (Sales from the highest sales city)
- **Total Sales per Factories** (Sales from the highest factories)
- **Product with the highest sales** (Highest product sales)
- **Shipping Distance** (Average shipping distance from the factory to the city with the highest sales)
- **Sales per Division** (Measures sales division from sales)
- **Unit Sold** (Measures how much total unit sold)
- **Gross Profit per Sales** (Measures the total of gross profits from sales)

Dashboard



Dashboard US Candy Sales

Filter

Total of Sales

122.95K

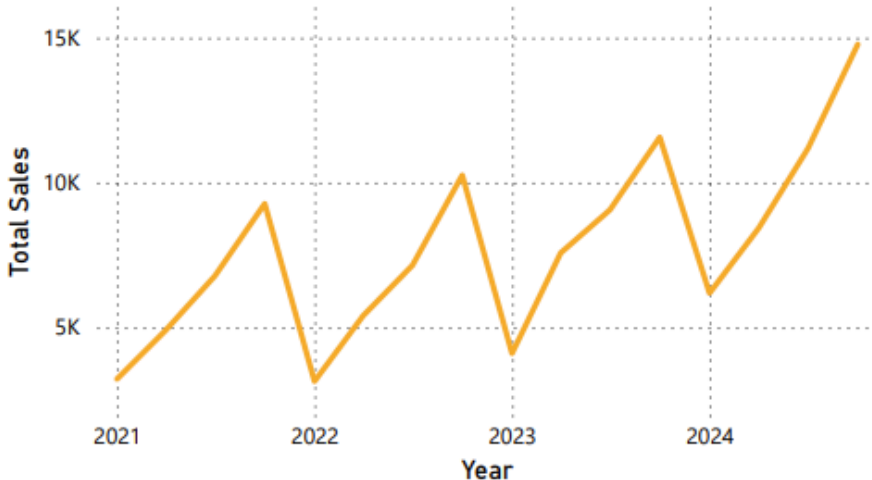
Total of Gross Profit

82.14K

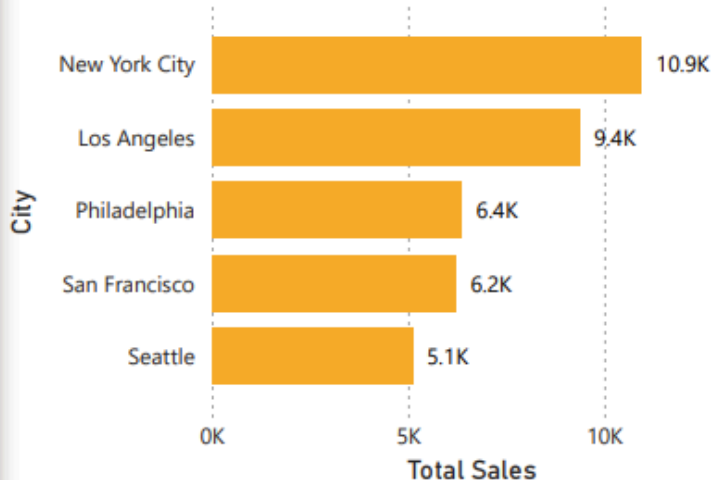
Total of Units

35K

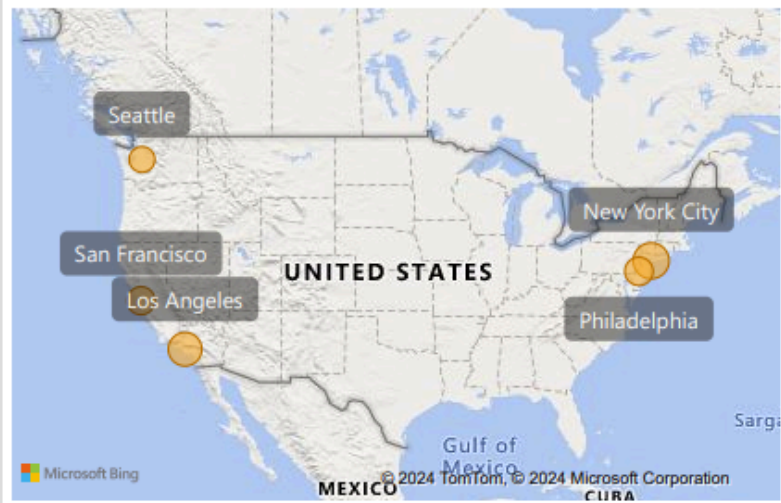
Total Sales by Year and Quarter



Total Sales by City



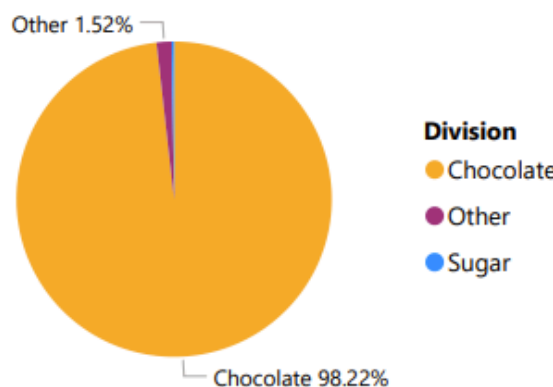
Comparison distance total sales by City



Top Sales by Product



Total Sales by Division



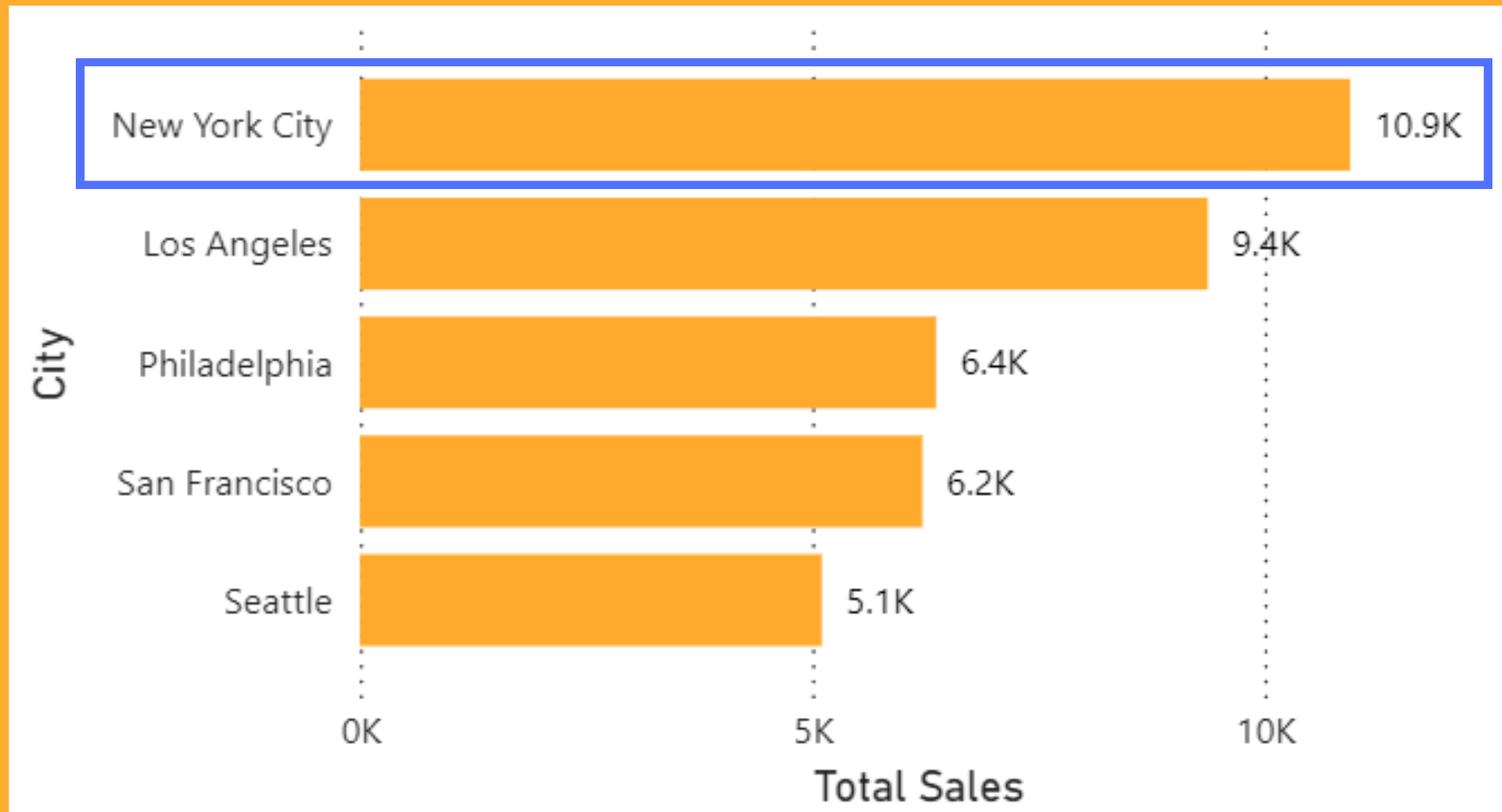
Average distance between City and Factory

City	Factory	Average of Distance
Los Angeles	Lot's O' Nuts	10,904.20
Los Angeles	Secret Factory	8,960.87
Los Angeles	The Other Factory	8,934.63
Los Angeles	Wicked Choccy's	8,036.00
New York City	Lot's O' Nuts	11,102.11
New York City	Secret Factory	9,143.10
New York City	Sugar Shack	259.47
New York City	The Other Factory	9,191.40
New York City	Wicked Choccy's	8,237.21
Philadelphia	Lot's O' Nuts	11,096.08
Philadelphia	Secret Factory	9,138.84
Philadelphia	Sugar Shack	249.53
Philadelphia	The Other Factory	9,178.41

Analysis

- **New York City** emerges as the city with the **highest sales** contribution, **totaling \$10.9K**, followed by **Los Angeles (\$9.4K)**, **Philadelphia (\$6.4K)**, **San Francisco (\$6.2K)**, and **Seattle (\$5.1K)**.
- This indicates that major cities in the US serve as key markets, with **New York City** standing out as the **dominant market hub**.

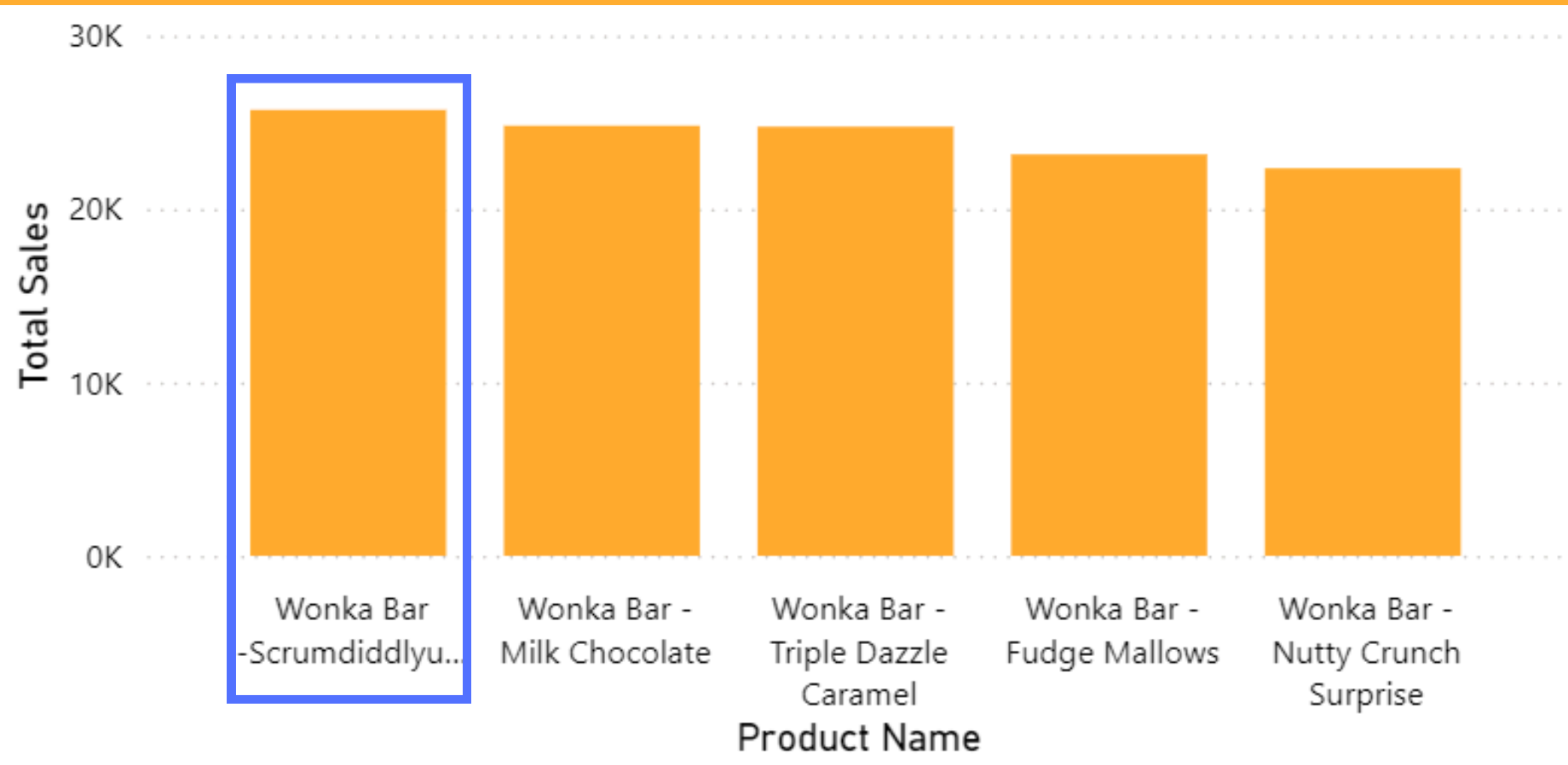
TOTAL SALES BY CITY



Analysis

Flagship products such as the Wonka Bar in **Scrumdiddlyumptious** and **Milk Chocolate** variants, with sales of **\$25.7K** and **\$24.8K** respectively, **dominate the list of best-selling products.**

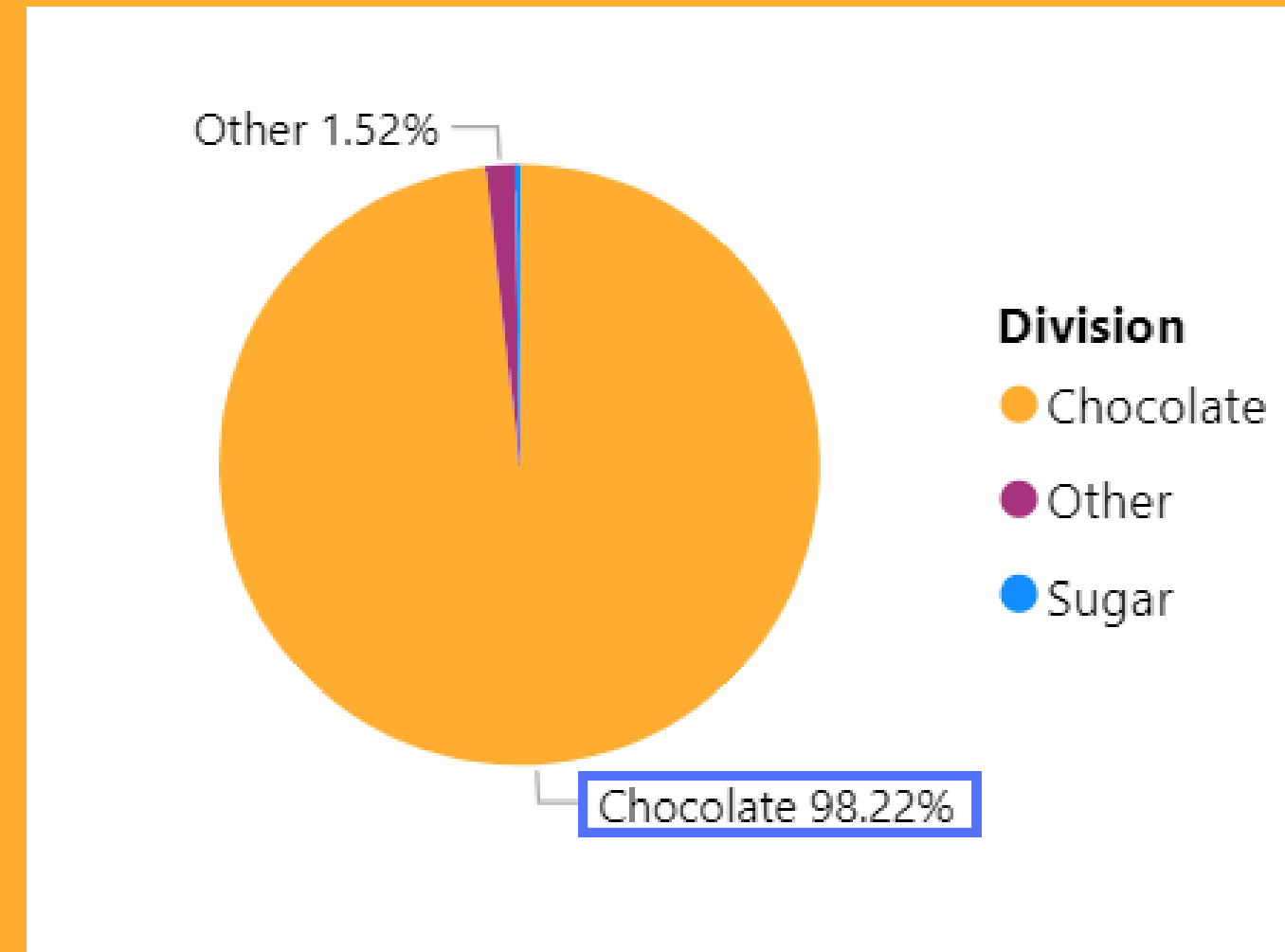
TOP SALES PRODUCT



Analysis

- Nearly **99% of sales** come from the **Chocolate category**, while the contributions of **Other** and **Sugar** categories are minimal, accounting for only **1.52%** and **0.26%**, respectively.
- This highlights that the **business focus on Chocolate products** has a **significant impact on overall sales**.

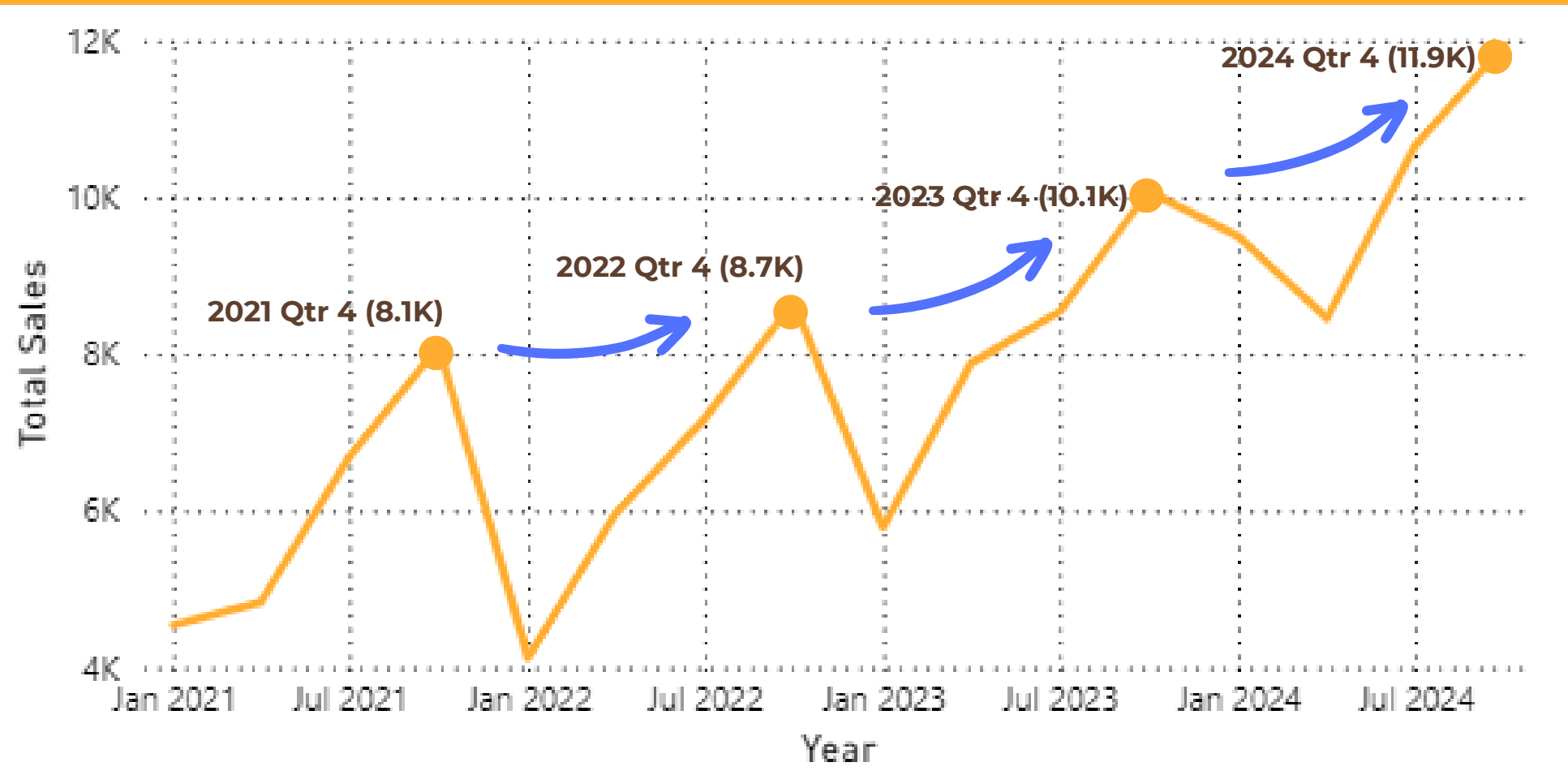
TOTAL SALES BY DIVISION



Analysis

- Sales have shown a **significant upward trend** since **2021**, rising from approximately **\$4K** to **\$12K** in **2024**.
- The sharp increase is particularly evident in **quarter 4**, reflecting the impact of **effective distribution and marketing strategies**.

TOTAL SALES BY YEAR & QUARTER



Analysis

- New York City**, with the highest total sales, has the **longest average distribution distance** to all factories, which is about **9.8K km**.
- Seattle** has a **closer average distribution distance** of **9.4K km**.

AVERAGE DISTANCE BETWEEN CITY & FACTORY

City	Factory	Average of Distance
Seattle	Lot's O' Nuts	10,703.28
Seattle	Secret Factory	8,846.66
Seattle	The Other Factory	8,653.95
Seattle	Wicked Choccy's	7,837.51
San Francisco	Lot's O' Nuts	10,673.91
San Francisco	Secret Factory	8,760.45
San Francisco	Sugar Shack	312.98
San Francisco	The Other Factory	8,652.70
San Francisco	Wicked Choccy's	7,800.40
Philadelphia	Lot's O' Nuts	11,096.08
Philadelphia	Secret Factory	9,138.84
Philadelphia	Sugar Shack	249.53
Philadelphia	The Other Factory	9,178.41
Philadelphia	Wicked Choccy's	8,230.66
New York City	Lot's O' Nuts	11,102.11
New York City	Secret Factory	9,143.10
New York City	Sugar Shack	259.47
New York City	The Other Factory	9,191.40
New York City	Wicked Choccy's	8,237.21
Los Angeles	Lot's O' Nuts	10,904.20
Los Angeles	Secret Factory	8,960.87
Los Angeles	The Other Factory	8,934.63
Los Angeles	Wicked Choccy's	8,036.00

Conclusions

- **New York City (\$10.9K)** has the **highest sales**, so this means customers there are buying a lot. We want to focus on other 5 cities too to boost their sales.
- **Wonka Bar - Scrumdiddlyumptious (\$25.7K)** has the **most sales** is probably the most popular. We want promote this more to increase overall sales.
- Sales are higher at every **quarter 4 of the year**, maybe because of **halloween, special promotions, and holidays**. Knowing this helps plan for future sales events and stocking up at the right time.



Recommendations

Recomendation 1 - Sales Strategy for New York City



Production and Distribution Program

- Increase the production and distribution of **Wonka Bar - Scumdiddlyumptious** to **the Wicked Choccy's factory (8.2K km)**, which is closer to **New York City** compared to **Lot's O' Nuts (11.1K km)**.
- **Scumdiddlyumptious** has the **highest sales** in **New York City** over **\$2.7K**, presenting a significant opportunity to maintain market dominance.

Impact

A more **efficient distribution strategy** will enhance product availability in key markets, driving further **sales growth in New York City**.

Recommendations

Recomendation 2 - Increasing Sales in Seattle



Optimizing Product Distribution

- Allocate **Wonka Bar - Scrumdiddlyumptious** to the **Wicked Choccy's factory (7.8K km)**, which is closer to **Seattle** compared to **Lot's O' Nuts (10.7K km)**.
- Currently, **Seattle** ranks **lowest among the top 5 cities** in terms of sales, with **Triple Dazzle Caramel (\$1.1K)** and **Scrumdiddlyumptious (\$1.08K)** as the two best-selling products.

Impact

This strategy will enhance the **penetration of Scrumdiddlyumptious** in **Seattle**, leading to a **gradual increase in sales**.

Recommendations

Recomendation 3 - Discount Program for the Top 5 Sales Cities



Flagship Product Promotion

Implement a **5-10%** discount on the best-selling products in each city:

- **New York City:** Wonka Bar - Scrumdiddlyumptious (\$2.7K)
- **Los Angeles:** Wonka Bar - Milk Chocolate (\$2.1K)
- **Philadelphia:** Wonka Bar - Fudge Mallows (\$1.3K)
- **San Francisco:** Wonka Bar - Triple Dazzle Caramel (\$1.4K)
- **Seattle:** Wonka Bar - Triple Dazzle Caramel (\$1.1K)

Impact

This initiative aims to **boost consumer appeal** and **accelerate sales growth** in the top-performing cities, with a combined contribution of **\$38K.43 to total sales**.

Recommendations

Recomendation 4 - Increasing Sales of Less Popular Products



Product Bundling Strategy

Focus on marketing **Kazookles** and **Wonka Gum**, which currently have low total sales (**\$590.5**), by creating bundling packages with its top products in the top 5 sales cities:

- **New York City:** Bundled with **Scrumdiddlyumptious**
- **Los Angeles:** Bundled with **Milk Chocolate**
- **Philadelphia:** Bundled with **Fudge Mallows**
- **San Francisco & Seattle:** Bundled with **Triple Dazzle Caramel**

Impact

This strategy aims to **enhance awareness and appeal** for underperforming products by leveraging the strategic markets of cities with the **highest sales contributions**.



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Thank You

Prepared and presented by: Team 7 Section Sabang

Appendix



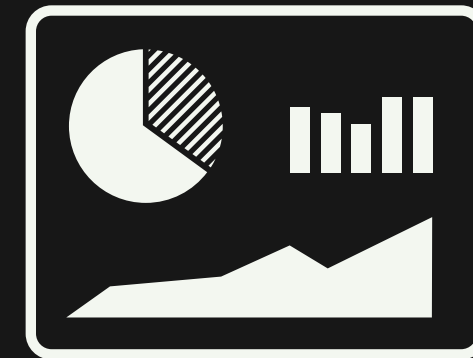
[Dataset Link](#)



[Business Analysis](#)



[Cleaned Data](#)



[Dashboard](#)